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2025-2-CISC3003-001 Web Programming

Taste of Macau: A Visual Food Finder

Team 04

Title Page

Reporters

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PRESENTED BY: **WONG POU I**

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Project Context

This is a full-stack web development project for CISC3003. It combines:

Front-End	Back-End
HTML5	PHP
CSS3	SQLite
JavaScript	Session Management

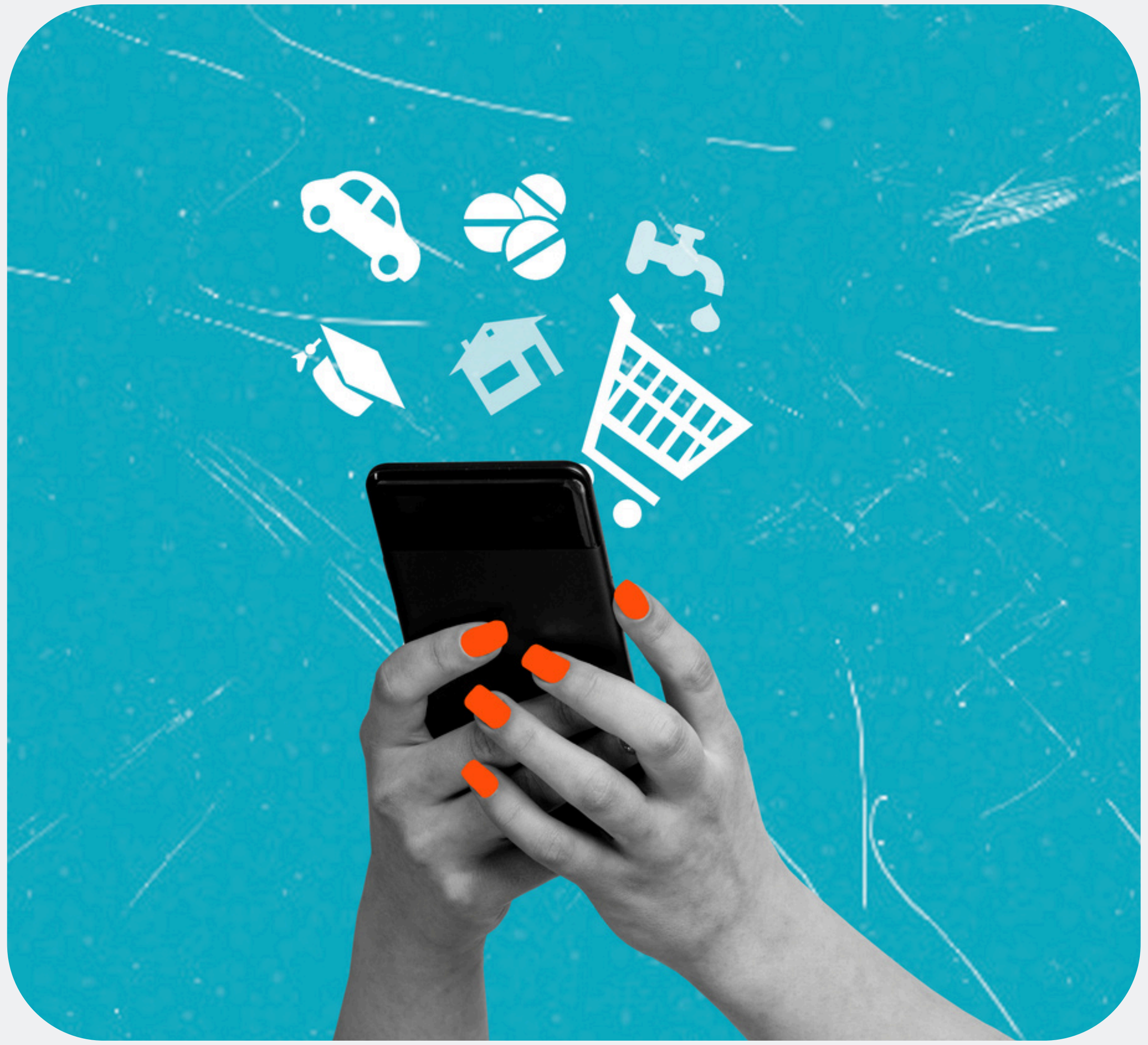
GenAI tools will assist with content generation and code debugging.
Trilingual architecture (zh/en/pt) with dual i18n system.
Purpose: Showcase Macau's culinary heritage through a visual, map-integrated platform.

Project Context

Precisely Written Question

How can an interactive, visually driven web application be designed to effectively catalogue and present information about Macau's diverse food culture, allowing users to browse and discover dishes based on categories and location, while providing complete user account management (signup/email verification/login/password reset), a personal user dashboard, and e-commerce style features (search history, browsing history, and favorites)?

Project Goals



The primary goal is to design an interactive, visually driven web application that effectively catalogues and presents information about Macau's diverse food culture.

It aims to provide a complete platform where users can browse and discover dishes by category and location, manage their accounts with full authentication, save favorites that sync from guest to registered accounts, explore detailed restaurant pages with user reviews and star ratings, and access a personal dashboard with search history, browsing history, and favorites — all available in Macau's three official languages: Chinese, English, and Portuguese.

Project Services and Data Requirements

Services

- User Authentication (Signup, Email Verification, Login, Password Reset)
- Category Filtering (Street Snacks, Portuguese Dishes, Desserts, Main Dishes)
- Search Functionality (by name, category, location)
- Interactive Map Integration with Dual-Mode Rendering (Food Markers + Restaurant Clusters) (Leaflet.js)
- Restaurant Detail Pages – 16 restaurants with signature dishes, reviews, embedded maps
- Review System – CRUD, 1–5 star ratings, automatic rating aggregation
- Favorites System – guest localStorage + server sync on login
- User Dashboard – Profile info, Search History, Browsing History (food + restaurant), Favorites
- Admin Panel – restaurant/food CRUD, feedback triage
- Session Management
- Trilingual Support (zh/en/pt with AJAX-persisted locale)
- Guest User Handling (LocalStorage)
- Feedback Mechanism – users report location issues or data inaccuracies → admin review pipeline

Project Services and Data Requirements

Data Requirements

- Food Food Dataset: 45-50 iconic Macau dishes selected by cultural representativeness, geographical distribution, and popularity, with images, descriptions, categories, and location coordinates verified through a three-layer mechanism
- Restaurant Dataset: 16 Macau restaurants with trilingual details, ratings, coordinates
- User Database: Tables for user accounts (id, username, email, hashed password, verification tokens, reset tokens)
- Activity Database: Tables for logging search queries, viewed food items, and viewed restaurant items – with deduplication
- Favorites Table – user-food relationships
- Reviews Table – user ratings + comments linked to restaurants
- User Feedback Table – location issue reports with status workflow (new → reviewing → resolved)

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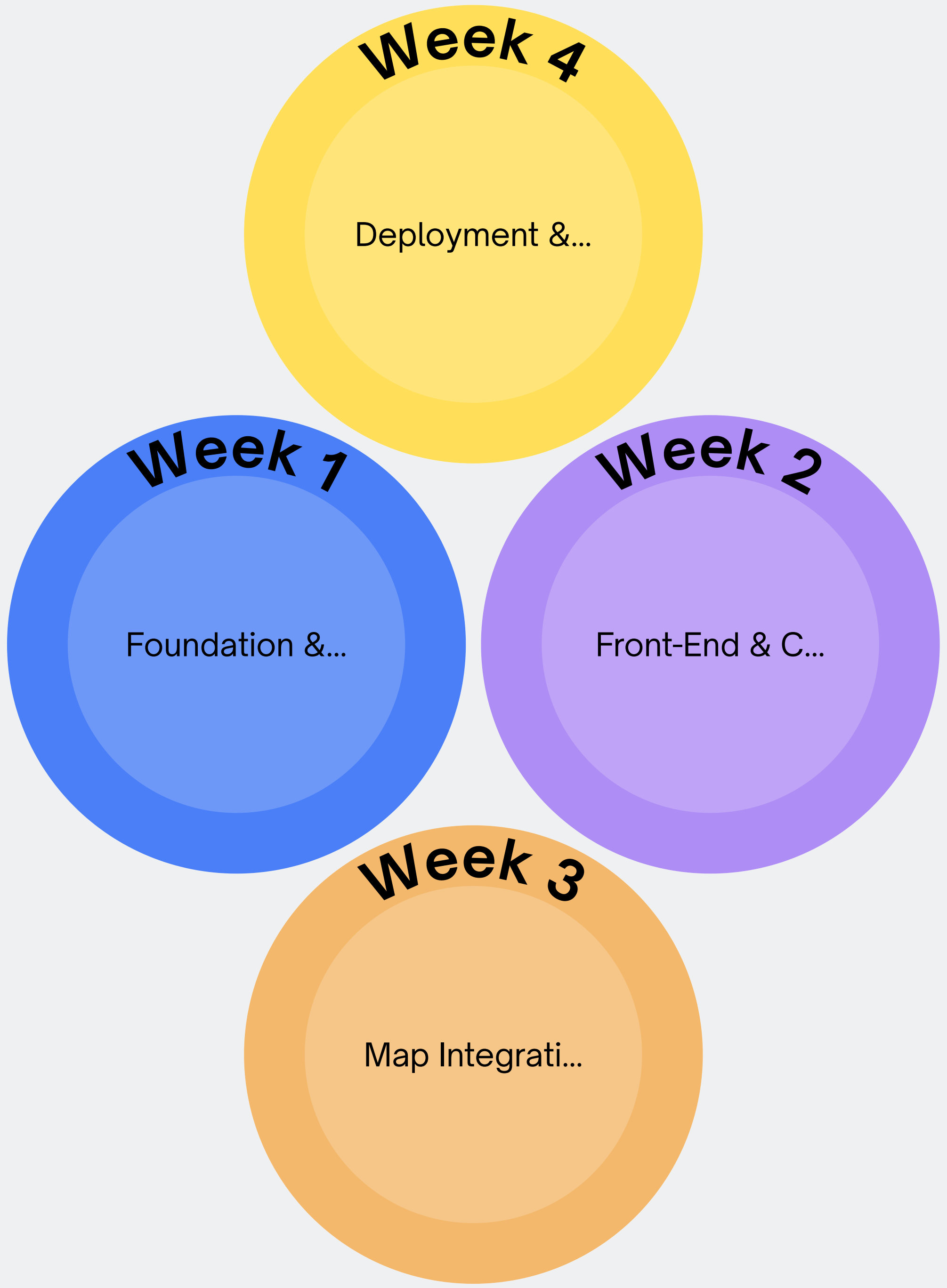
Project Tasks

- Week 1 - Foundation & User Authentication: Curated 49 dishes with verified locations, designed the SQLite schema (12 tables), and built the PHP authentication system (register/login/verify/reset) on MVC architecture.
- Week 2 - Front-End & Core Features: Built responsive Macau-themed UI, restaurant and food catalogs with debounced search and filters, favorites system, and guest-to-user data merge.
- Week 3 - Map Integration & Dashboard: Integrated Leaflet map with dual-mode rendering (food markers + restaurant clusters), bi-directional card-marker linking, review system with star ratings, and user dashboard.
- Week 4 - Deployment & Testing: Built restaurant detail pages and admin panel, finalized back-end integration, deployed to Apache server at food.jackie-macau.top, and completed cross-browser testing.



Project Challenges

- Integrating Leaflet.js and establishing bi-directional interactivity between food cards, restaurant cards, and map markers (dual-mode rendering), including marker clustering behavior
- Handling the edge case where a restaurant has multiple signature dishes at the same coordinates
- Implementing a secure user authentication flow (email verification, password reset) with token hashing, expiration windows, and CSRF protection across SSR form submissions and AJAX API calls

Project Challenges

- Designing a database that logs user activity and supports guest-to-login data merging, using INSERT OR IGNORE semantics and client-side deduplication
- Deploying a PHP application with SQLite on a live Apache server – mod_rewrite, .htaccess, WAL mode, file permissions
- Managing trilingual content pipeline – every entry needed translations across zh/en/pt, and dual i18n systems (server + client) that stay in sync
- Managing guest user data via LocalStorage with secure merge logic – no PII collected, only anonymous favorites and browsing activity

Project Deliverables

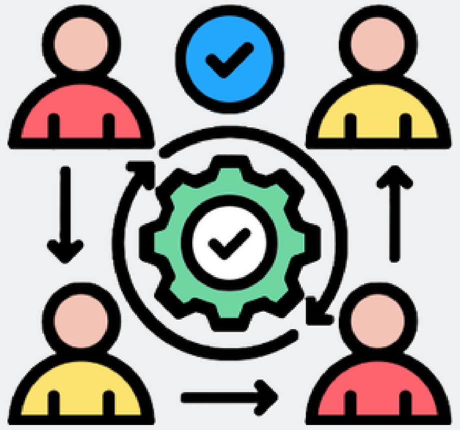
01 **Working authentication system + SQLite database + food selection & location verification plan (45-50 dishes)**

02 **Responsive, trilingual front-end with category filters, full-text search, and favorites – LocalStorage for guests, DB for registered users**

03 **Interactive map with dual-mode rendering (food markers + restaurant clusters), bi-directional card-marker linking, plus dashboard with search history, browsing history, and favorites**

04 **Individual restaurant detail pages with signature dishes, full user review system with star ratings and automatic rating aggregation, and admin panel for content management**

05 **Live URL at https://food.jackie-macau.top – fully tested full-stack app with user feedback mechanism for reporting data issues**



Project Division of Work Among Team Members

	Role	Responsibility
Jiang Jin Hong	**Front-End Design**	HTML, CSS, responsive layout, visual elements
Leong Chi Long	**JavaScript Logic**	Filters, search, LocalStorage, guest merge
Li Wu Yue	**Map Integration**	Leaflet, marker clustering, card-map linking
Wong Pou I	**PHP Back-End**	Authentication, sessions, email, form handling
Wu Ze Xian	**Database & Dashboard**	SQL schema, history logging, dashboard UI
Zhang Zi Han	**Deployment & QA**	Hosting, testing, debugging, documentation

PRESENTED BY: **WU ZE XIAN**

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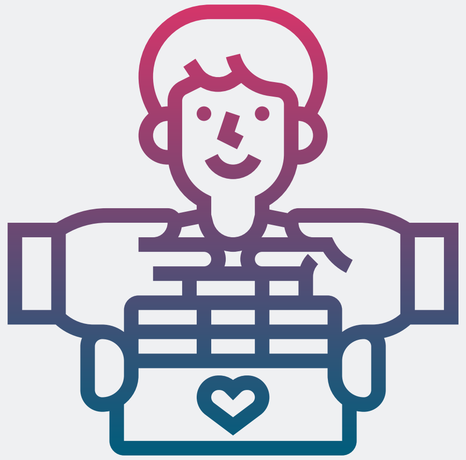
Project Contributions From Team Members



Members	Key Contribution Areas
Jiang Jin Hong	Responsive UI design, food card rendering, mobile-first layout, visual styling
Leong Chi Long	Category filtering, search bar logic, guest LocalStorage handling, login merge flow
Li Wu Yue	Interactive map setup, marker clustering (Leaflet.markercluster), bi-directional card-marker interaction
Wong Pou I	User signup/login/password reset, email verification, session management, PHP form processing
Wu Ze Xian	SQLite schema design, search/browsing history storage, dashboard data retrieval, feedback table
Zhang Zi Han	Deployment, cross-browser testing, responsive debugging, final documentation

PRESENTED BY: **WU ZE XIAN**

Criteria of Success

- Testable Responsive Design
- Visual Appeal
- Full-Stack Implementation
- User Authentication Success
- User Dashboard Success
- Map Interactivity & Clustering Success
- Review System Success
- Guest User Handling Success
- Trilingual Coverage
- Successful Deployment
- Code Quality

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Project Evaluation

Evaluation will be based on a self-assessment checklist aligned with the success criteria.

The definitive measure will be the live demonstration of the web application at the public URL, which will be used by the instructor to assess team and individual contributions against the course learning outcomes.

Our application is deployed on our own Apache server at https://food.jackie-macau.top, where every feature — from user registration to map interaction to the review system — can be tested in real time.

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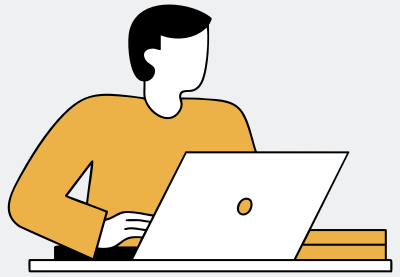
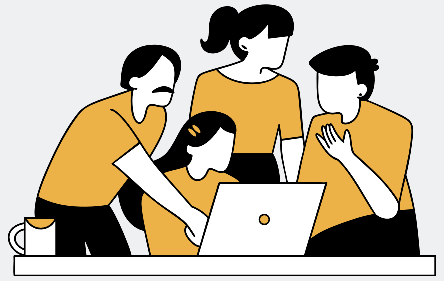
Project Learning - Individual + Team



Individual Learning

- DOM manipulation & asynchronous JavaScript
- PHP session management, password hashing (password_hash/verify), CSRF protection patterns
- Geospatial data integration (Leaflet) & dual-mode rendering (marker clustering + individual markers), mobile-responsive map
- SQLite schema design with foreign keys, cascading deletes, WAL-mode concurrency
- Full-stack deployment on Apache with mod_rewrite, SSL configuration, PHP production settings
- Guest data management (LocalStorage & merge logic)
- Trilingual application architecture – database column suffixes, dual i18n systems (server + client), locale persistence



Team Learning

- Collaboration on a complex full-stack project using a systematic development methodology.
- Application of information architecture principles for digital presentation of cultural heritage (Macau Gastronomy).
- Integration of multiple services (search, maps, user history, feedback mechanism) into a cohesive single-page application experience.
- Managing custom MVC framework complexity without relying on Laravel, Composer, or npm – giving deep understanding of how frameworks actually work under the hood.

Project Declaration of AI Usage

AI tools supported our development:

Claude Opus 4.7

(Anthropic)

A flagship model known for deep reasoning, high-quality code generation, and safety-oriented design.

Project Declaration of AI Usage

How We Used It?

1. Code Development & Debugging

- Assisted with map clustering bugs, async/await refactoring, SQL injection fixes
- Treated as a debugging partner: paste error logs, analyze suggestions, test before integrating

2. Code Review & Cross-File Comprehension

- Mapping PHP controller → JS module interactions

3. Database Architecture

- Helped brainstorm normalization strategies for activity logging tables
- Final architectural decisions made by the team

4. Documentation & Presentation

- Assisted with structuring this presentation's narrative flow

Project Declaration of AI Usage

Boundaries & Reflection:

What AI Did NOT Do:

- Core authentication logic → Wong Pou I
- CSS responsive grid design → Jiang Jin Hong
- Deployment configuration → Zhang Zi Han

Our Principle:

- AI served as an accelerator and sounding board, never a replacement.
- Every AI-suggested line was reviewed, understood, and tested before commit.

What We Learned:

AI dramatically speeds up development

AI can confidently generate errors – human oversight is essential

Verifying AI output is now a core competency for web developers

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Thank You

Any questions?

Team 04

https://food.jackie-macau.top